

Final System Proof — Boba Merge Service (live nezha @ e800ebd)

Revision: 1 **Last modified:** 2026-06-17T00:00:00Z **Authority:** End-to-end live verification (read-only) against the deployed stack **Scope:** Full-fleet ASCII + multi-word + Cyrillic searches; encoding-crash census; private-tracker auth; gamestorrents

Headline

Three full-fleet searches (29-tracker fleet) ran sequentially against the live deployed merge service on nezha.local:7187 (HTTP 200 confirmed before testing). All three returned large real result sets. The P0 system-wide query-encoding fix holds for the core trackers — Cyrillic Война и мир returned correctly-decoded Cyrillic titles from rutracker/nnmclub/rutor. Private-tracker authentication is live (rutracker, nnmclub, kinozal, iptorrents all authenticated=true). gamestorrents is fixed (returns results on relevant queries).

Encoding-crash census across all 3 queries: ONE plugin_bad_query_encoding — the yts plugin only, only on the Cyrillic query. This is an isolated per-plugin defect (yts builds its URL without URL-encoding), NOT the P0 system-level crash; it does not affect the core/private trackers. Reported honestly below.

Query results (live, async POST /api/v1/search → poll GET /api/v1/search/{id})

Query	Type	total_results	merged_results	contributing trackers (>0)	plugin_bad_query_encoding
ubuntu	ASCII	1578	489	21	0
the matrix	multi-word	2395	1127	23	0
Война и мир	Cyrillic (P0)	543	373	11	1 (yts)

Query echoed back by the API on the Cyrillic POST as Война и мир — no mojibake at the request boundary.

Per-key-tracker status

Tracker	ubuntu	the matrix	Война и мир	auth
rutracker	50	50	50	✓ authenticated
nmclub	48	50	50	✓ authenticated
kinozal	0 (empty)	50	0 (empty)	✓ authenticated
iptorrents	49	49	0 (empty)	✓ authenticated
gamestorrents	0 (empty)	6	0 (empty)	n/a (public)

All four private trackers report `authenticated=true` on every query — `auth` is live. Empty cells are genuine no-match results (e.g. `iptorrents/kinozal` have no “Война и мир” matches), not `auth` failures or crashes.

Cyrillic correctness proof (real decoded titles, P0)

From `Война и мир` results — `rutracker` returned correctly-decoded Cyrillic: - `Война и мир (Сергей Бондарчук) [1965–1968, СССР, драма, ...]` - `Толстой Л. – Война и мир (иллюстрации А.П. Апсита и Н.Н.Каразина) [2021, ...]` - `Толстой Лев – Война и Мир [Некрасов Денис, 2008, 128 kbps, MP3]`

This is the direct end-user proof the P0 encoding fix works for the core trackers.

gamestorrents

Fixed — returned 6 results on the `matrix` (relevant game/film title). Empty on `ubuntu` / `Война и мир` (genuinely no matches), so the plugin runs cleanly (status empty, no crash). No scoped GTA re-run needed: the `matrix` already proves `results_count > 0`.

Honest notes (§11.4.6 — facts, not bluff)

- **yts plugin_bad_query_encoding on Cyrillic** (`Война и мир`): the `yts` plugin built `/browse-movies/Война и мир/all/...` WITHOUT URL-encoding, so `urllib` rejected it (URL can't contain control characters ... found ' '). `yts` SUCCEEDED on the `matrix` (22 results), so this is a Cyrillic-specific `yts`-only defect — a per-plugin bug, NOT the P0 system crash. Core/private trackers are unaffected. This is a real residual finding; the task’s “expect NONE” was not fully met for the `yts` plugin on the Cyrillic query. UPDATE (post-test, same session): a sibling fix to `plugins/yts.py` — percent-encoding each `browse-movies` path segment via `quote(..., safe='')` — landed in the working tree with a dedicated RED→GREEN test (`tests/unit/test_plugin_unicode_query_encoding.py::test_yts_browse_path_cyrillic_encoded`, 44/44 GREEN, §1.1 mutation confirmed). It is NOT yet committed or deployed to the `e800ebd` live stack, so the numbers above reflect the live system as tested. Re-test after deploy to confirm the `yts` Cyrillic crash clears.
- **snowfl plugin_crashed on ubuntu**: `stderr Connection error: Service Unavailable + non-JSON body → JSON-parse raise`. Transient upstream flake,

NOT an encoding crash. snowfl SUCCEEDED on the matrix (140 results), confirming it is upstream availability, not a code defect.

- **deadline_timeout** (tokyotoshokan, yourbittorrent, torlock on some queries) is a slow-upstream cutoff, NOT a crash — the tracker simply did not respond in the deadline window.
- **upstream_http_403** (kickass, yts-on-some) and **upstream_http_404** (megapeer on Cyrillic) are upstream HTTP responses, NOT plugin crashes.

Method / environment

- Read-only against live stack: `ssh milosvasic@nezha.local 'curl localhost:7187/...'`
- Sequential searches (one at a time), async POST + poll GET to be gentle on the concurrent service.
- Proxy (7186) and merge service (7187) both HTTP 200 before testing.
- Raw evidence JSON captured under `/Volumes/T7/tmp/{ubuntu,matrix_final,cyr_final}.json`.

Post-yts-fix re-verify (c05423e deployed)

After deploying the yts browse-path UTF-8 fix (c05423e), the Cyrillic full-fleet Война и мир re-ran on nezha: **total=608, 29 trackers, encoding-crashed/plugin_crashed: NONE**. yts no longer emits `plugin_bad_query_encoding` — it now returns `upstream_http_403` (yts.mx is Cloudflare-gated, an UPSTREAM block like kickass §11.4.112, not a code defect). Net: **zero encoding crashes across the entire fleet on ASCII, multi-word, AND Cyrillic** — the residual is closed.